9 (a) Write down the value of $36^{\frac{1}{2}}$

A power of one half means square root: √36 = 6.

1 mark
Principal square root

6

(1)

(b) Write down the value of 23^0

Any non-zero number raised to the power 0 is 1.

1 mark
Apply the zero-index rule

1

(1)

(c) Work out the value of $27^{-\frac{2}{3}}$

The cube root of 27 is 3, since 3 × 3 × 3 = 27.

Square this: $3^2 = 9$. A negative power takes the reciprocal.

1 mark
Use a cube root or a reciprocal

1 mark
Exact fraction

Answer = $\frac{1}{9}$ $\frac{1}{9}$

(2)

(Total for Question 9 is 4 marks)

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